# Formulario de examen · Estadística Matemática

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Estadística I (repaso) + Est Mat · FCNM–ESPOL

**Parametrizaciones:** el Formulario 2 usa $\beta$ como **escala** ($=\mathbb{E}$) en Exp/Gamma. La "exponencial de parámetro $\lambda$" lleva $\lambda=$ **tasa**: $\beta=1/\lambda$, media $1/\lambda$. $N(\mu,\sigma^2)$ lleva la **varianza**; Geo/Bneg cuentan **ensayos**.

**Modelos (parametrización del Formulario 2; si son independientes las FGM se multiplican)**

| **Modelo** | $\mathbb{P}(X=x)$ **o** $f_X(x)$ | **FDA** $F_X(x)$ $\lfloor\cdot\rfloor$ ent.; $I_x$ beta; $P$ gamma | $\mathbb{E}(X)$ | $\mathrm{Var}(X)$ | $M_X(t)$ |
|---|---|---|---|---|---|
| $\mathrm{Ber}(p)$ | $p^x(1-p)^{1-x},\ x\in\{0,1\}$ | $0\ (x<0);\ 1-p\ (0\le x<1);\ 1\ (x\ge1)$ | $p$ | $p(1-p)$ | $pe^t+1-p$ |
| $\mathrm{Bin}(n,p)$ | $\binom{n}{x}p^x(1-p)^{n-x},\ x=0,\dots,n$ | $\sum_{k=0}^{\lfloor x\rfloor}\binom{n}{k}p^k(1-p)^{n-k}$ | $np$ | $np(1-p)$ | $(pe^t+1-p)^n$ |
| $\mathrm{Geo}(p)$ | $p(1-p)^{x-1},\ x=1,2,\dots$ | $1-(1-p)^{\lfloor x\rfloor}\ (x\ge1);\ 0\ (x<1)$ | $1/p$ | $(1-p)/p^2$ | $\dfrac{pe^t}{1-(1-p)e^t}$ |
| $\mathrm{Bneg}(r,p)$ | $\binom{x-1}{r-1}p^r(1-p)^{x-r},\ x\ge r$ | $\sum_{k=r}^{\lfloor x\rfloor}\binom{k-1}{r-1}p^r(1-p)^{k-r}$ | $r/p$ | $r(1-p)/p^2$ | $\Big[\dfrac{pe^t}{1-(1-p)e^t}\Big]^r$ |
| $\mathrm{Hiper}(N,a,n)$ | $\binom{a}{x}\binom{N-a}{n-x}/\binom{N}{n}$ | $\sum_{k=0}^{\lfloor x\rfloor}\dfrac{\binom{a}{k}\binom{N-a}{n-k}}{\binom{N}{n}}$ | $n\frac{a}{N}$ | $n\frac{a}{N}(1-\frac{a}{N})\frac{N-n}{N-1}$ | sin forma cerrada |
| $\mathrm{Pois}(\lambda)$ | $e^{-\lambda}\lambda^x/x!,\ x=0,1,\dots$ | $e^{-\lambda}\sum_{k=0}^{\lfloor x\rfloor}\lambda^k/k!$ | $\lambda$ | $\lambda$ | $e^{\lambda(e^t-1)}$ |
| Unif. discreta | $1/k,\ x\in\{x_1,\dots,x_k\}$ | $\frac{1}{k}\#\{i:x_i\le x\}$ | $\frac{1}{k}\sum x_i$ | $\frac{1}{k}\sum x_i^2-\bar{x}^2$ | $\frac{1}{k}\sum_i e^{tx_i}$ |
| $U(a,b)$ | $\frac{1}{b-a},\ a<x<b$ | $0\ (x<a);\ \frac{x-a}{b-a}\ (a\le x<b);\ 1\ (x\ge b)$ | $\frac{a+b}{2}$ | $\frac{(b-a)^2}{12}$ | $\dfrac{e^{tb}-e^{ta}}{t(b-a)}$ |
| $N(\mu,\sigma^2)$ | $\frac{1}{\sqrt{2\pi\sigma^2}}e^{-(x-\mu)^2/(2\sigma^2)}$ | $\Phi(\frac{x-\mu}{\sigma})$ | $\mu$ | $\sigma^2$ | $e^{\mu t+\sigma^2t^2/2}$ |
| $\mathrm{Exp}(\beta)$ | $\frac{1}{\beta}e^{-x/\beta},\ x>0$ | $1-e^{-x/\beta}\ (x\ge0);\ 0\ (x<0)$ | $\beta$ | $\beta^2$ | $(1-\beta t)^{-1}$ |
| $G(\alpha,\beta)$ | $\frac{1}{\beta^\alpha\Gamma(\alpha)}x^{\alpha-1}e^{-x/\beta},\ x>0$ | $P(\alpha,\frac{x}{\beta})$ | $\alpha\beta$ | $\alpha\beta^2$ | $(1-\beta t)^{-\alpha}$ |
| $\chi^2_{(\nu)}=G(\frac{\nu}{2},2)$ | $\frac{1}{2^{\nu/2}\Gamma(\nu/2)}x^{\nu/2-1}e^{-x/2}$ | $P(\frac{\nu}{2},\frac{x}{2})$ | $\nu$ | $2\nu$ | $(1-2t)^{-\nu/2}$ |
| $\mathrm{Beta}(\alpha,\beta)$ | $\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)},\ 0<x<1$ | $I_x(\alpha,\beta)$ | $\frac{\alpha}{\alpha+\beta}$ | $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ | $\mathbb{E}(X^k)=\frac{B(\alpha+k,\beta)}{B(\alpha,\beta)}$ |

**Dominios FGM:** Geo/Bneg: $t<-\ln(1-p)$; Exp/Gamma: $t<1/\beta$; $\chi^2$: $t<\frac{1}{2}$; $U$: $t\neq0$.

**Relaciones (por FGM o definición):** $\mathrm{Ber}_n$ i.i.d. $\to\mathrm{Bin}(n,p)$; Geo=Bneg$(1,p)$; Bneg=suma de $r$ Geo i.i.d.; $\mathrm{Pois}(\lambda_i)$ indep. $\to\mathrm{Pois}(\sum\lambda_i)$; $\mathrm{Exp}(\beta)=G(1,\beta)$; $n$ Exp$(\beta)$ i.i.d. $\to G(n,\beta)$; $G(\alpha_i,\beta)$ (mismo $\beta$) $\to G(\sum\alpha_i,\beta)$; $N(\mu_i,\sigma_i^2)$ indep. $\to N(\sum\mu_i,\sum\sigma_i^2)$; $aX+b\sim N(a\mu+b,a^2\sigma^2)$; $Z^2\sim\chi^2_{(1)}$; $\sum^\nu\chi^2_{(1)}\to\chi^2_{(\nu)}$; $\chi^2_{(2)}=$Exp(2); $\mathrm{Beta}(1,1)=U(0,1)$.

## 1. Variable aleatoria, densidad y acumulada

Discreta: $p_X(x)=\mathbb{P}(X=x)\ge0$, $\sum_{S_X}p_X=1$, $\mathbb{P}(X\in A)=\sum_A p_X$.

Continua: $f_X\ge0$, $\int_{-\infty}^{\infty}f_X=1$, $\mathbb{P}(a<X\le b)=\int_a^b f_X$.

$$F_X(x)=\mathbb{P}(X\le x),\quad \mathbb{P}(a<X\le b)=F_X(b)-F_X(a).$$

Continua: $F_X(x)=\int_{-\infty}^{x}f_X(t)\,dt$, $f_X=F_X'$. Discreta: $\mathbb{P}(X=x)=F(x)-F(x^-)$. **Cuantil** $p$**:** $x_p$ con $F_X(x_p)=p$ (mediana $x_{0.5}$); si $F$ creciente, $x_p=F^{-1}(p)$. **Ojo:** $f_X$ *no* es probabilidad (puede pasar de 1); "$<$" vs "$\le$" da igual en continuas, no en discretas.

## 2. Esperanza, varianza, covarianza

$$\mathbb{E}(X)=\sum_x x\,\mathbb{P}(X=x)=\int x f_X(x)\,dx,$$

$$\mathbb{E}[g(X)]=\sum_x g(x)\mathbb{P}(X=x)=\int g(x)f_X(x)\,dx.$$

$$\mathrm{Var}(X)=\mathbb{E}\big[(X-\mathbb{E}X)^2\big]=\mathbb{E}(X^2)-[\mathbb{E}X]^2,\quad \sigma_X=\sqrt{\mathrm{Var}(X)}.$$

$$\mathrm{Cov}(X,Y)=\mathbb{E}(XY)-\mathbb{E}(X)\mathbb{E}(Y)=\mathbb{E}\big[(X-\mathbb{E}X)(Y-\mathbb{E}Y)\big].$$

- **E:** $\mathbb{E}(aX+b)=a\mathbb{E}X+b$; $\mathbb{E}(X+Y)=\mathbb{E}X+\mathbb{E}Y$ (siempre); $\mathbb{E}[\sum a_iX_i+b]=\sum a_i\mathbb{E}X_i+b$; $\mathbb{E}(c)=c$.
- $X\perp\!\!\!\perp Y\Rightarrow\mathbb{E}(XY)=\mathbb{E}X\,\mathbb{E}Y$ y $\mathbb{E}[g(X)h(Y)]=\mathbb{E}g(X)\mathbb{E}h(Y)$.
- $\mathbb{E}[g(X)]\neq g(\mathbb{E}X)$ salvo $g$ lineal; $g$ convexa $\Rightarrow\mathbb{E}g(X)\ge g(\mathbb{E}X)$ (Jensen). **Esperanza total:** $\mathbb{E}X=\mathbb{E}[\mathbb{E}(X\mid Y)]$.
- **Var:** $\mathrm{Var}(aX+b)=a^2\,\mathrm{Var}\,X$; $\mathrm{Var}(X+b)=\mathrm{Var}\,X$; $\mathrm{Var}\,c=0$.
- $\mathrm{Var}(X\pm Y)=\mathrm{Var}\,X+\mathrm{Var}\,Y\pm2\,\mathrm{Cov}(X,Y)$; $\mathrm{Var}(\sum X_i)=\sum\mathrm{Var}\,X_i+2\sum_{i<j}\mathrm{Cov}(X_i,X_j)$; si indep. $\mathrm{Var}(\sum X_i)=\sum\mathrm{Var}\,X_i$.
- **Varianza total:** $\mathrm{Var}\,X=\mathbb{E}[\mathrm{Var}(X\mid Y)]+\mathrm{Var}(\mathbb{E}[X\mid Y])$.
- **Cov:** $\mathrm{Cov}(X,Y)=\mathrm{Cov}(Y,X)$; $\mathrm{Cov}(X,X)=\mathrm{Var}\,X$; $\mathrm{Cov}(aX+b,cY+d)=ac\,\mathrm{Cov}(X,Y)$; $\mathrm{Cov}(X+Y,Z)=\mathrm{Cov}(X,Z)+\mathrm{Cov}(Y,Z)$; $\mathrm{Cov}(\sum a_iX_i,\sum b_jY_j)=\sum_i\sum_j a_ib_j\,\mathrm{Cov}(X_i,Y_j)$.
- $X\perp\!\!\!\perp Y\Rightarrow\mathrm{Cov}=0$ (recíproco sólo en la normal bivariada). $\rho=\mathrm{Corr}=\frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}\,X\,\mathrm{Var}\,Y}}\in[-1,1]$, $\mathrm{Corr}(aX+b,cY+d)=\mathrm{sign}(ac)\rho$.
- Cauchy–Schwarz $[\mathrm{Cov}(X,Y)]^2\le\mathrm{Var}\,X\,\mathrm{Var}\,Y$. **Markov** ($X\ge0$): $\mathbb{P}(X\ge a)\le\mathbb{E}X/a$.
- **Chebyshev:** $\mathbb{P}(|X-\mu|\ge k)\le\sigma^2/k^2$; en $k\sigma$: $\le1/k^2$.

## 3. Función generadora de momentos (FGM)

$$M_X(t)=\mathbb{E}(e^{tX}),\quad M_X(0)=1,$$

$$M_X^{(k)}(0)=\mathbb{E}(X^k),\quad \mathrm{Var}\,X=M_X''(0)-[M_X'(0)]^2.$$

- $M_{aX+b}(t)=e^{bt}M_X(at)$.
- $X_i\perp\!\!\!\perp\Rightarrow M_{\sum X_i}(t)=\prod_i M_{X_i}(t)$; i.i.d. $\Rightarrow[M_X(t)]^n$.
- **Unicidad:** si $M_X=M_Y$ cerca de 0, misma distribución. *Clave del método FGM.*

**Receta:** FGM de cada sumando (tabla) $\to$ multiplicar (usa $M_{a+bX}$) $\to$ simplificar a una FGM conocida $\to$ identificar y concluir por unicidad.

## 4. Gamma y Beta (truco del núcleo)

$$\Gamma(\alpha)=\int_0^\infty x^{\alpha-1}e^{-x}dx,\quad \alpha>0;$$

$$\Gamma(\alpha+1)=\alpha\Gamma(\alpha),\quad \Gamma(n)=(n-1)!,\quad \Gamma(\tfrac{1}{2})=\sqrt{\pi}.$$

$$B(\alpha,\beta)=\int_0^1 x^{\alpha-1}(1-x)^{\beta-1}dx=\frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$$

**Núcleo:** $\int_0^\infty x^{\alpha-1}e^{-x/\beta}dx=\beta^\alpha\Gamma(\alpha)$, $\int_0^1x^{\alpha-1}(1-x)^{\beta-1}dx=B(\alpha,\beta)$, $\int e^{-\frac{(x-\mu)^2}{2\sigma^2}}dx=\sqrt{2\pi\sigma^2}$. **Momentos Gamma:** $\mathbb{E}(X^k)=\beta^k\frac{\Gamma(\alpha+k)}{\Gamma(\alpha)}$.

## 5. Distribuciones conjuntas

$$f_X(x)=\int f_{XY}(x,y)\,dy,\quad f_{X|Y}(x\mid y)=\frac{f_{XY}(x,y)}{f_Y(y)},$$

$$X\perp\!\!\!\perp Y\iff f_{XY}=f_Xf_Y.$$

$$\mathbb{E}[g(X,Y)]=\iint g\,f_{XY}\,dx\,dy,\quad \mathbb{E}[X\mid Y=y]=\int x f_{X|Y}dx,$$

$$\mathbb{E}[\mathbb{E}(X\mid Y)]=\mathbb{E}X.$$

Discretas: cambia $\int$ por $\sum$.

- Soporte **no rectangular** (p. ej. $0<x<y<1$) $\Rightarrow$ no son independientes.
- Soporte rectangular y $f_{XY}=g(x)h(y)\Rightarrow$ sí independientes.
- **Límites:** dibuja el soporte; la variable de afuera tiene límites constantes; la de adentro, fija la de afuera y mira dónde entra/sale; si la frontera cambia, parte la integral.

## 6. Transformaciones (Clase 1)

**Acumulada (general):** (1) soporte de $U=h(Y)$; (2) $F_U(u)=\mathbb{P}(h(Y)\le u)$; (3) despeja $Y$ (si $h$ decrece, voltea); (4) usa $F_Y$; (5) $f_U=F_U'$.

$$\text{Jacobiano 1D (monótona):}\quad f_U(u)=f_Y[h^{-1}(u)]\Big|\frac{dh^{-1}}{du}\Big|.$$

$$\text{No monótona:}\quad f_U(u)=\sum_k f_Y[h_k^{-1}(u)]\Big|\frac{dh_k^{-1}}{du}\Big|,$$

$$U=Y^2:\quad f_U(u)=\frac{f_Y(\sqrt{u})+f_Y(-\sqrt{u})}{2\sqrt{u}}\ (u>0).$$

$$\text{Bivariado:}\quad f_{U_1,U_2}=f_{Y_1,Y_2}(h_1^{-1},h_2^{-1})\,|J|,\quad J=\det\big[\partial y_i/\partial u_j\big];$$

$$n\text{-D:}\quad g(\mathbf{u}) = f(\mathbf{y}(\mathbf{u}))\left|\det\frac{\partial \mathbf{y}}{\partial \mathbf{u}}\right|.$$

$J$ es el jacobiano de la **inversa**; atajo $J = 1/\det(\partial(u_1,u_2)/\partial(y_1,y_2))$. Si piden una sola $U_1$, añade una **auxiliar** y marginaliza. Tramos planos $\Rightarrow$ átomos: usa sólo la acumulada.

## 7. Estadísticos de orden (Clase 2)

$Y_{(1)}$ = mín, $Y_{(n)}$ = máx de $Y_i$ i.i.d.

$$F_{(n)}(y) = [F(y)]^n,\quad g_{(n)}(y) = n[F(y)]^{n-1}f(y);$$
$$F_{(1)}(y) = 1-[1-F(y)]^n,\quad g_{(1)}(y) = n[1-F(y)]^{n-1}f(y).$$

$$\text{Teo. 1:}\quad f_{Y_{(i)}}(x) = \frac{n!}{(i-1)!(n-i)!}[F(x)]^{i-1}f(x)[1-F(x)]^{n-i}.$$

$$f_{(1),\dots,(n)} = n!\prod_k f(y_k);$$
$$f_{(i),(j)}(x,y) = c_{ij}\,f(x)f(y)\,F(x)^{i-1}[F(y)-F(x)]^{j-i-1}[1-F(y)]^{n-j},$$
$$c_{ij} = \frac{n!}{(i-1)!(j-i-1)!(n-j)!}\ (i<j).$$

Mín. de Exp($\beta$) i.i.d. $\sim$ Exp($\beta/n$); $Y_{(i)}$ de $U(0,1)\sim$ Beta$(i, n-i+1)$.

## 8. Distribuciones muestrales: $\chi^2$ y $t$ (Clase 3)

$Y_1,\dots,Y_n$ i.i.d. $N(\mu,\sigma^2)$; $\bar Y = \frac{1}{n}\sum Y_i$, $S^2 = \frac{1}{n-1}\sum(Y_i-\bar Y)^2$.

$$\bar Y \sim N\Big(\mu,\frac{\sigma^2}{n}\Big)\ \Leftrightarrow\ Z = \frac{\bar Y-\mu}{\sigma/\sqrt{n}} \sim N(0,1);$$
$$\sum_i\Big(\frac{Y_i-\mu}{\sigma}\Big)^2 \sim \chi^2_{(n)}.$$

$$\bar Y \perp\!\!\!\perp S^2,\quad \frac{(n-1)S^2}{\sigma^2}\sim\chi^2_{(n-1)};$$
$$\chi^2_{(\nu)} = G(\tfrac{\nu}{2},2):\ \mathbb{E}=\nu,\ \mathrm{Var}=2\nu,\ M=(1-2t)^{-\nu/2}.$$

$$Z\sim N(0,1)\perp\!\!\!\perp W\sim\chi^2_{(\nu)}\Rightarrow T=\frac{Z}{\sqrt{W/\nu}}\sim t_{(\nu)};$$
$$\nu=n-1:\quad \frac{\bar Y-\mu}{S/\sqrt{n}}\sim t_{(n-1)}.$$

$t_{(\nu)}$: $\mathbb{E}=0\,(\nu>1)$, $\mathrm{Var}=\frac{\nu}{\nu-2}\,(\nu>2)$, $\to N(0,1)$ si $\nu\to\infty$. FDA: $F_t(t) = 1-\frac{1}{2}I_{\nu/(\nu+t^2)}(\frac{\nu}{2},\frac{1}{2})$ $(t\ge 0)$.

## 9. La distribución $F$ (Clase 4)

$$W_1\sim\chi^2_{(\nu_1)}\perp\!\!\!\perp W_2\sim\chi^2_{(\nu_2)}\Rightarrow F=\frac{W_1/\nu_1}{W_2/\nu_2}\sim F_{(\nu_1,\nu_2)};$$
$$\mathbb{E}=\frac{\nu_2}{\nu_2-2}\,(\nu_2>2),\quad \mathrm{Var}=\frac{2\nu_2^2(\nu_1+\nu_2-2)}{\nu_1(\nu_2-2)^2(\nu_2-4)}\,(\nu_2>4).$$

FDA: $F(x) = I_{\nu_1 x/(\nu_1 x+\nu_2)}\big(\frac{\nu_1}{2},\frac{\nu_2}{2}\big)$; $I$ = beta incompleta regularizada.

- $X\sim t_{(\nu)}\Rightarrow X^2\sim F_{(1,\nu)}$, y $F_{(1,\nu,\alpha)} = t^2_{(\nu),\alpha/2}$.
- $X\sim F_{(\nu_1,\nu_2)}\Rightarrow 1/X\sim F_{(\nu_2,\nu_1)}$; $\quad F_{(\nu_1,\nu_2,1-\alpha)} = 1/F_{(\nu_2,\nu_1,\alpha)}$.
- Muestras normales indep.: $\dfrac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}\sim F_{(n_1-1,n_2-1)}$; si $\sigma_1^2=\sigma_2^2$, $S_1^2/S_2^2\sim F_{(n_1-1,n_2-1)}$.

## 10. Teorema del límite central (Clase 5)

$Y_i$ i.i.d., $\mathbb{E}Y_i=\mu$, $\mathrm{Var}=\sigma^2<\infty$; $S_n=\sum Y_i$:

$$W_n = \frac{\bar Y-\mu}{\sigma/\sqrt{n}} = \frac{S_n-n\mu}{\sigma\sqrt{n}}\ \xrightarrow{d}\ N(0,1);$$
$$\bar Y\overset{a}{\sim}N\Big(\mu,\frac{\sigma^2}{n}\Big),\quad S_n\overset{a}{\sim}N(n\mu,n\sigma^2).$$

Bernoulli: Bin$(n,p)\overset{a}{\sim}N(np,np(1-p))$ si $np\ge5$ y $n(1-p)\ge5$. Sin $\sigma^2$ finita no hay TLC.

- **Corrección $\pm\frac{1}{2}$:** $X\le k\Rightarrow\Phi(\frac{k+0.5-\mu_X}{\sigma_X})$; $X<k\Rightarrow\Phi(\frac{k-0.5-\mu_X}{\sigma_X})$; $X\ge k\Rightarrow 1-\Phi(\frac{k-0.5-\mu_X}{\sigma_X})$; $X>k\Rightarrow 1-\Phi(\frac{k+0.5-\mu_X}{\sigma_X})$.
- $X=k\ \Rightarrow\ \Phi(\frac{k+0.5-\mu_X}{\sigma_X})-\Phi(\frac{k-0.5-\mu_X}{\sigma_X})$; $a\le X\le b\ \Rightarrow\ \Phi(\frac{b+0.5-\mu_X}{\sigma_X})-\Phi(\frac{a-0.5-\mu_X}{\sigma_X})$.

## 11. Estimación puntual y ECM (Clase 6)

Insesgado: $\mathbb{E}[\hat\theta]=\theta\iff B(\hat\theta)=\mathbb{E}[\hat\theta]-\theta=0$. Descomposición MSE: $\mathrm{MSE}_\theta(\hat\theta)=\mathbb{E}[(\hat\theta-\theta)^2]=\mathrm{Var}(\hat\theta)+[B(\hat\theta)]^2$; si insesgado, MSE = Var. En $N(\mu,\sigma^2)$: $\bar X$ insesgado (Var $=\frac{\sigma^2}{n}$); cuasivarianza $S^2$ insesgada (Var $=\frac{2\sigma^4}{n-1}$); varianza $S_n^2$ sesgada con $B=-\frac{\sigma^2}{n}<0$; Bessel: $S^2=\frac{n}{n-1}S_n^2$. Con dos datos: $\frac{(X_1-X_2)^2}{2}$ insesgado (Var $=2\sigma^4$).

## 12. Cota de error y tamaño de muestra (Clase 7)

Asintóticamente insesgado: $\lim_{n\to\infty}B(\hat\theta_n)=0$ ($S_n^2$ sí lo es; $T_n=(\bar X)^2-\bar X$ en Poisson no). Cota $b$ ($\approx 95\,\%$): $b=2\sigma_{\hat\theta}$ o $z_{\alpha/2}\sigma_{\hat\theta}$. Errores estándar: $\sigma_{\bar Y}=\frac{\sigma}{\sqrt{n}}$; $\sigma_{\hat p}=\sqrt{\frac{p(1-p)}{n}}\le\frac{1}{2\sqrt{n}}$; $\sigma_{\bar Y_1-\bar Y_2}=\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}$; $\sigma_{\hat p_1-\hat p_2}=\sqrt{\frac{p_1(1-p_1)}{n_1}+\frac{p_2(1-p_2)}{n_2}}$. Tamaño de muestra: para $\mu$, $n=\big(\frac{2\sigma}{b}\big)^2$ ($n\propto 1/b^2$); para $p$, $n=\frac{4p(1-p)}{b^2}\le\frac{1}{b^2}$ (conservador en $p=0.5$).

## 13. Tablas y valores críticos

Las tablas ESPOL dan área a la **izquierda** $q$: $\mathbb{P}(X\le x)=q$. **Crítico con cola derecha $\alpha$ = tabla en columna $q=1-\alpha$.** El percentil $100p$ es $x_p$; el crítico al $5\,\%$ es el percentil 95.

- $Z$: $\Phi(z_\alpha)=1-\alpha$, $\Phi(-z)=1-\Phi(z)$.
- $\chi^2_{(\nu),\alpha}$: columna $1-\alpha$. $\quad t_{(\nu),\alpha}$: columna $1-\alpha$; $\mathbb{P}(|T|\le c)=2\mathbb{P}(T\le c)-1$ (simetría).
- $F_{(\nu_1,\nu_2)}$: bloque $\nu_2$ (denominador), $F_{(\nu_1,\nu_2,\alpha)}$ = fila $q=1-\alpha$, columna $\nu_1$ (numerador).